

Semantic Bridge: IBM 3090 Channel Subsystem (CSS)

Technical Expansion

1. CCW (Channel Command Word) Mapping Logic

The SemanticBridge translates modern asset metadata into Mainframe Format 1 CCWs. Each CCW consists of a Command Code, Data Address, Flags, and Count.

CCW Mapping Table

Asset Attribute	CCW Field	Mainframe Command	Description
read_request	Command Code	0x02 (READ)	Transfers data from device to storage
write_data	Command Code	0x01 (WRITE)	Transfers data from storage to device
sense_id	Command Code	0xE4 (SENSE ID)	Identifies the device type (Hardware Hub v2, etc).
payload_ptr	Data Address	31-bit Address	Memory pointer for data buffer
data_len	Count	16-bit Count	Number of bytes to transfer

2. EBCDIC Translation Layer

Since mainframes natively use EBCDIC (Extended Binary Coded Decimal Interchange Code), the bridge implements a high-performance bitwise translation layer for JSON strings.

Python Translation Stub

```
def to_ebcdic(utf8_string):
    # Mapping table for UTF-8 to EBCDIC Code Page 037
    return utf8_string.encode('cp037')

def from_ebcdic(ebcdic_bytes):
    return ebcdic_bytes.decode('cp037')
```

3. Subchannel (SCHIB) Initialization

For the SymbioStream assets to be recognized by the MVS/ESA or VM/XA operating systems, the bridge must populate the Subchannel Information Block (SCHIB).

Path Management Control Word (PMCW): Configures the CHPID (Channel Path ID) to point to the virtualized SymbioStream hardware hub.

Interrupt Parameter: Set to 0x53594D (SYM) for identification during I/O interrupts.

4. Operation Request Block (ORB) Generation

The bridge generates an ORB to initiate the SSCH (Start Subchannel) instruction

```
struct orb {
    uint32_t int_parm;    // 0x53594D
    uint32_t key_flags;   // Storage key and control flags
    uint32_t ccw_addr;    // Pointer to the first CCW in the chain
    // ... additional control fields
};
```